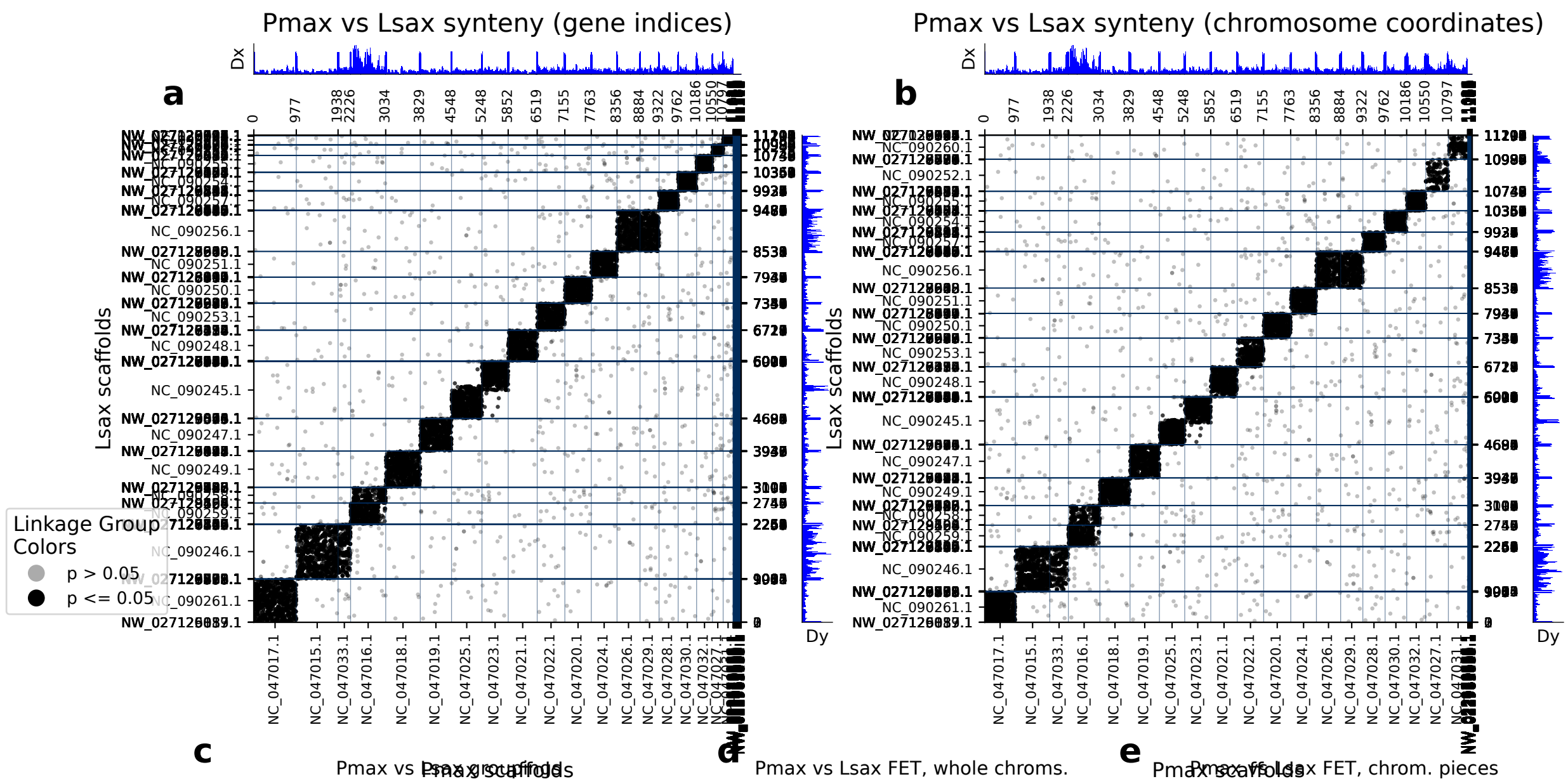


diamond: 11213 orthologs on 192 Pmax scaffolds and 232 Lsax scaffolds.



	Pmax scaf	Lsax scaf	p (FET)	Count
0	NC_047017.1	NC_090261.1	0.00e+00	903
1	NC_047015.1	NC_090246.1	0.00e+00	889
2	NC_047018.1	NC_090249.1	0.00e+00	741
3	NC_047019.1	NC_090247.1	0.00e+00	660
4	NC_047025.1	NC_090245.1	0.00e+00	632
5	NC_047021.1	NC_090248.1	0.00e+00	627
6	NC_047023.1	NC_090245.1	0.00e+00	546
7	NC_047022.1	NC_090253.1	0.00e+00	539
8	NC_047024.1	NC_090251.1	0.00e+00	529
9	NC_047020.1	NC_090250.1	0.00e+00	528
10	NC_047026.1	NC_090256.1	0.00e+00	466
11	NC_047016.1	NC_090259.1	0.00e+00	424
12	NC_047029.1	NC_090256.1	0.00e+00	381
13	NC_047028.1	NC_090257.1	0.00e+00	379
14	NC_047030.1	NC_090254.1	0.00e+00	361
15	NC_047032.1	NC_090255.1	0.00e+00	310
16	NC_047016.1	NC_090258.1	6.52e-286	290
17	NC_047033.1	NC_090246.1	6.48e-194	247
18	NC_047027.1	NC_090252.1	7.35e-294	184
19	NC_047031.1	NC_090260.1	9.71e-239	152
20	NW_022980724.1	NC_030595.1	1.90e-07	3
21	NW_022980858.1	NW_027125817.1	1.28e-02	2

C	Pmax scaf	LSax scaf	p (FET)	Count
0	NC_047017.1 : 0-end	NC_090261.1 : 0-end	0.00e+00903	
1	NC_047015.1 : 0-end	NC_090246.1 : 0-end	0.00e+00889	
2	NC_047018.1 : 0-end	NC_090249.1 : 0-end	0.00e+00741	
3	NC_047019.1 : 0-end	NC_090247.1 : 0-end	0.00e+00660	
4	NC_047025.1 : 0-end	NC_090245.1 : 0-end	0.00e+00632	
5	NC_047021.1 : 0-end	NC_090248.1 : 0-end	0.00e+00627	
6	NC_047023.1 : 0-end	NC_090245.1 : 0-end	0.00e+00546	
7	NC_047022.1 : 0-end	NC_090253.1 : 0-end	0.00e+00539	
8	NC_047024.1 : 0-end	NC_090251.1 : 0-end	0.00e+00529	
9	NC_047020.1 : 0-end	NC_090250.1 : 0-end	0.00e+00528	
10	NC_047026.1 : 0-end	NC_090256.1 : 0-end	0.00e+00466	
11	NC_047016.1 : 0-end	NC_090259.1 : 0-end	0.00e+00424	
12	NC_047029.1 : 0-end	NC_090256.1 : 0-end	0.00e+00381	
13	NC_047028.1 : 0-end	NC_090257.1 : 0-end	0.00e+00379	
14	NC_047030.1 : 0-end	NC_090254.1 : 0-end	0.00e+00361	
15	NC_047032.1 : 0-end	NC_090255.1 : 0-end	0.00e+00310	
16	NC_047016.1 : 0-end	NC_090258.1 : 0-end	6.52e-286290	
17	NC_047033.1 : 0-end	NC_090246.1 : 0-end	6.48e-194247	
18	NC_047027.1 : 0-end	NC_090252.1 : 0-end	7.35e-294184	
19	NC_047031.1 : 0-end	NC_090260.1 : 0-end	9.71e-239152	
20	NW_022980724.1 : 0-end	NC_030595.1 : 0-end	1.90e-07 3	
21	NW_022980858.1 : 0-end	NW_027125817.1 : 0-end	1.28e-02 2	

(a) depicts the chromosomes/scaffolds of Pmax (x-axis) plotted against the chromosomes/scaffolds of Lsax (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 11213 orthologs found between 192 Pmax scaffolds and 232 Lsax scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. *Nature Ecology & Evolution* 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chromosome FET p-value, and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.